Homework 9

This homework is due at 11 PM on April 3, 2026.

Submission Format: Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned), as well as a printout of your completed Jupyter notebook(s).

1. **Optimizing Over Multiple Variables**

   In this exercise, we consider several problems in which we optimize over two variables, $\vec{x} \in \mathbb{R}^n$ and $\vec{y} \in \mathbb{R}^m$, and a general (possibly nonconvex) objective function, $F_0(\vec{x}, \vec{y})$. Suppose also that $\vec{x}$ and $\vec{y}$ are constrained to different feasible sets $\mathcal{X}$ and $\mathcal{Y}$, respectively, which may or may not be convex.

   (a) Show that

   $$\min_{\vec{x} \in \mathcal{X}} \min_{\vec{y} \in \mathcal{Y}} F_0(\vec{x}, \vec{y}) = \min_{\vec{y} \in \mathcal{Y}} \min_{\vec{x} \in \mathcal{X}} F_0(\vec{x}, \vec{y}), \tag{1}$$

   i.e., if we minimize over both $\vec{x}$ and $\vec{y}$, then we can exchange the minimization order without altering the optimal value.

   (b) Show that $p^\star \geq d^\star$, where

   $$p^\star \doteq \min_{\vec{x} \in \mathcal{X}} \max_{\vec{y} \in \mathcal{Y}} F_0(\vec{x}, \vec{y}) \tag{2}$$

   $$d^\star \doteq \max_{\vec{y} \in \mathcal{Y}} \min_{\vec{x} \in \mathcal{X}} F_0(\vec{x}, \vec{y}). \tag{3}$$

   This statement is referred to as the *min-max theorem*.

2. Duality

Consider the function

$$f(\vec{x}) = \vec{x}^\top A\vec{x} - 2\vec{b}^\top \vec{x}. \tag{4}$$

First, we consider the unconstrained optimization problem

$$p^\star = \min_{\vec{x}\in\mathbb{R}^n} f(\vec{x}) = \min_{\vec{x}\in\mathbb{R}^n} \vec{x}^\top A\vec{x} - 2\vec{b}^\top \vec{x} \tag{5}$$

for a real $n \times n$ symmetric matrix $A \in \mathbb{S}^n$ and $\vec{b} \in \mathbb{R}^n$. If the problem is unbounded below, then we say $p^\star = -\infty$. Let $\vec{x}^\star$ denote the minimizing argument of the optimization problem.

(a) Suppose $A \succeq 0$ (positive semidefinite) and $\vec{b} \in \mathcal{R}(A)$. Let $\operatorname{rank}(A) = n$. Find $p^\star$.

HINT: What does $A \succeq 0$ tell you about the function f? How can you leverage the rank of A to compute $p^\star$?

(b) Suppose $A \succeq 0$ (positive semidefinite) and $\vec{b} \in \mathcal{R}(A)$ as before. Let A be rank-deficient, i.e., $\operatorname{rank}(A) = r < n$. Let A have the compact/thin and full SVD as follows, with diagonal positive definite $\Lambda_r \in \mathbb{R}^{r\times r}$:

$$A = U_r\Lambda_r U_r^\top = \begin{bmatrix} U_r & U_1 \end{bmatrix} \begin{bmatrix} \Lambda_r & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} U_r^\top \\ U_1^\top \end{bmatrix}. \tag{6}$$

Show that the minimizer $\vec{x}^\star$ of the optimization problem (5) is not unique by finding a general form for the family of solutions for $\vec{x}^\star$ in terms of $U_r, U_1, \Lambda_r, \vec{b}$.

HINT: As before, $A \succeq 0$ gives you some information about the objective function f. Can you use this information along with the fact that $b \in \mathcal{R}(A)$ to obtain a general form for the minimizers of f? Use the fact that any vector $\vec{x} \in \mathbb{R}^n$ can be written as $\vec{x} = U_r\vec{\alpha} + U_1\vec{\beta}$ for unique $\vec{\alpha}, \vec{\beta}$.

(c) If $A \not\succeq 0$ (A not positive semi-definite) show that $p^\star = -\infty$ by finding $\vec{v}$ such that $f(\alpha\vec{v}) \to -\infty$ as $\alpha \to \infty$.

HINT: $A \not\succeq 0$ means that there exists $\vec{v}$ such that $\vec{v}^\top A\vec{v} < 0$.

(d) Suppose $A \succeq 0$ (positive semidefinite) and $\vec{b} \notin \mathcal{R}(A)$. Find $p^\star$. Justify your answer mathematically.

HINT: From FTLA, we know that $\mathbb{R}^n = \mathcal{R}\left(A^\top\right) \oplus \mathcal{N}(A)$. Therefore, $\vec{b} = \vec{v}_1 + \vec{v}_2$ where $\vec{v}_1 \in \mathcal{R}(A) = \mathcal{R}\left(A^\top\right)$ and $\vec{v}_2 \in \mathcal{N}(A)$.

For parts 2(e) and 2(f), consider real $n \times n$ symmetric matrix $A \in \mathbb{S}^n$ and $\vec{b} \in \mathbb{R}^n$. Let $\operatorname{rank}(A) = r$, where $0 \le r \le n$. Now we consider the constrained optimization problem

$$\begin{aligned} p^\star = \min_{\vec{x}\in\mathbb{R}^n} \; & \vec{x}^\top A\vec{x} - 2\vec{b}^\top \vec{x} \\ \text{s.t. } & \vec{x}^\top \vec{x} \ge 1. \end{aligned} \tag{7}$$

(e) Write the Lagrangian $\mathcal{L}(\vec{x}, \lambda)$, where λ is the dual variable corresponding to the inequality constraint.

(f) For any matrix $C \in \mathbb{R}^{n\times n}$ with $\operatorname{rank}(C) = r \le n$ and compact SVD

$$C = U_r\Lambda_r V_r^\top, \tag{8}$$

we define the pseudoinverse as

$$C^\dagger = V_r\Lambda_r^{-1}U_r^\top. \tag{9}$$

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We use the "dagger" operator to represent this. If $\vec{d}$ lies in the range of C, then a solution to the equation $C\vec{x} = \vec{d}$, can be written as $\vec{x} = C^{\dagger}\vec{d}$, even when C is not full rank. Show that the dual problem to the primal problem (7) can be written as,

$$d^{\star} = \max_{\substack{\lambda \geq 0 \\ A - \lambda I \succeq 0 \\ \vec{b} \in \mathcal{R}(A - \lambda I)}} \quad -\vec{b}^{\top} (A - \lambda I)^{\dagger} \vec{b} + \lambda. \tag{10}$$

HINT: To show this, first argue that when the constraints are not satisfied $\min_{\vec{x}} \mathcal{L}(\vec{x}, \lambda) = -\infty$. *Then show that when the constraints are satisfied,* $\min_{\vec{x}} \mathcal{L}(\vec{x}, \lambda) = -\vec{b}^{\top} (A - \lambda I)^{\dagger} \vec{b} + \lambda$.

HINT: Compute $g(\lambda)$ *and explore its behavior under the constraints.*

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3. Minimizing a Sum of Logarithms

Consider the following problem, which arises in estimation of transition probabilities of a discrete-time Markov chain:

$$p^\star = \max_{\vec{x} \in \mathbb{R}^n} \quad \sum_{i=1}^{n} \alpha_i \log(x_i) \tag{11}$$

$$\text{s.t.} \quad \vec{x} \geq 0, \quad \vec{1}^\top \vec{x} = c, \tag{12}$$

where $c > 0$ and $\alpha_i > 0$, $i = 1, \ldots, n$. (Recall that if $\vec{x}$ is a vector then by "$\vec{x} \geq 0$" we mean "$x_i \geq 0$ for each i.") We will determine in closed-form a minimizer, and show that the optimal objective value of this problem is

$$p^\star = \alpha \log(c/\alpha) + \sum_{i=1}^{n} \alpha_i \log(\alpha_i), \tag{13}$$

where $\alpha \doteq \sum_{i=1}^{n} \alpha_i$. We will show this in a series of steps.

(a) First, express the problem as a minimization problem which has optimal value $p^\star_{\min}$.

(b) In optimization, we often "relax" problems of the form $p^\star_{\min} = \min_{\vec{x} \in \mathcal{X}} f_0(\vec{x})$, i.e., replacing the constraint set $\mathcal{X}$ with a larger constraint set $\mathcal{X}_r$, and instead solving $p^\star_r = \min_{\vec{x} \in \mathcal{X}_r} f_0(\vec{x})$, then showing a connection between $p^\star_{\min}$ and $p^\star_r$. In this problem, a particular relaxation we will use is to replace the equality constraint $\vec{1}^\top \vec{x} = c$ with an inequality constraint $\vec{1}^\top \vec{x} \leq c$.

Show that the relaxed problem has the same optimal value as the original problem, i.e., $p^\star_r = p^\star_{\min}$, and the two problems have the same solutions.

HINT: First argue that $p^\star_r \leq p^\star_{\min}$. Then, suppose for the sake of contradiction that $p^\star_r < p^\star_{\min}$. Let $\vec{x}^r$ be a solution to the relaxed minimization problem which has objective value $p^\star_r$. Consider the vector $\vec{x}$ given by

$$\vec{x} \doteq \begin{bmatrix} c - 1^\top \vec{x}^r + x_1^r \\ x_2^r \\ \vdots \\ x_n^r \end{bmatrix}. \tag{14}$$

Show that $\vec{x}$ is feasible for the original problem and has objective value $< p^\star_r$. Argue that this implies $p^\star_{\min} < p^\star_r$ and derive a contradiction. Finally, argue that any solution to the relaxed problem is a solution to the original problem, and vice-versa — you might need to use a construction similar to $\vec{x}$.

(c) After relaxing the equality constraint to an inequality constraint, form the Lagrangian $\mathcal{L}(\vec{x}, \vec{\lambda}, \mu)$ for the relaxed minimization problem, where λ_i is the dual variable corresponding to the inequality $x_i \geq 0$, and μ is the dual variable corresponding to the inequality constraint $\vec{1}^\top \vec{x} \leq c$.

(d) Now derive the dual function $g(\vec{\lambda}, \mu)$ for the relaxed minimization problem, and solve the dual problem $d^\star_r = \max_{\substack{\vec{\lambda} \geq \vec{0} \\ \mu \geq 0}} g(\vec{\lambda}, \mu)$. What are the optimal dual variables $\vec{\lambda}^\star, \mu^\star$?

(e) Show that strong duality holds for the relaxed problem, so $p^\star_r = d^\star_r$.

(f) From the $\vec{\lambda}^\star, \mu^\star$ obtained in the previous part, how do we obtain the optimal primal variable $\vec{x}^\star$? What is the optimal objective function value $p^\star_r$? Finally, what is $p^\star$?

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4. Lagrangian Dual of a QP

Consider the standard form of a convex quadratic program, with $Q \succ 0$:

$$\min_{\vec{x}} \quad \frac{1}{2}\vec{x}^\top Q\vec{x} \tag{15}$$

$$\text{s.t.} \quad A\vec{x} \leq \vec{b} \tag{16}$$

(a) Write the Lagrangian function $\mathcal{L}(\vec{x}, \vec{\lambda})$.

(b) Write the Lagrangian dual function, $g(\vec{\lambda})$.

(c) Show that the Lagrangian dual problem is convex by writing it in "standard QP form" – that is, in a form similar to the original problem. Is the Lagrangian dual problem convex in general?

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5. Dual of the dual of a linear program

Consider a standard linear program P:

$$\min_{\vec{x}} \quad \vec{c}^\top \vec{x} \tag{17}$$

$$\text{s.t.} \quad A\vec{x} = \vec{b} \tag{18}$$

$$\vec{x} \geq \vec{0}. \tag{19}$$

where $\vec{x}, \vec{c} \in \mathbb{R}^n, \vec{b} \in \mathbb{R}^m, A \in \mathbb{R}^{m \times n}$. $\vec{x} \geq \vec{0}$ means $x_i \geq 0$ for all $i = 1, \ldots, n$.

(a) Formulate the Lagrangian of the problem P, and write the dual problem.

Note: The dual problem should not have the variable $\vec{x}$.

(b) Express the dual problem as an equivalent minimization problem. Find the dual of this minimization problem, i.e., the dual of the dual. Compare it to the original linear program formulation.

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6. Homework Process

With whom did you work on this homework? List the names and SIDs of your group members.

NOTE: If you didn't work with anyone, you can put "none" as your answer.

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